Lam Fung Academy · LF Academy

Primary 5 · Lesson 28 · Student Handout

Comprehensive—Volume + Area + Fraction

Cross-Topic integration (T1 · T5 · T9) · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Units 7-8 + Modern Education 5B Unit 18-20

Core traps: 🔴 High Frequency

Sub-test cross-topic comprehensive questions, accounting for about 18-22% of Paper 1

Prerequisite knowledge: Class 21 (Circle + Section + Origami) · Class 22-23 (Fundamentals of Volume + Application) · Class 24-26 (Multiplication and Division of Fractions) · Class 27 (Division of Decimals)

Objectives of this class: ❶ Comprehensive application of volume and area formulas ❷ Application of fractions in volume ❸ Identification and avoidance of cross-topic traps

Our goals: ❶ Comprehensive application of volume and area formulas ❷ Application of fractions in volume ❸ Identification and avoidance of cross-topic traps

Student Name:	Class:	Date:	Time Spent:

I.Warm-Up Questions（total 5 question · 5 minutes）

#	Question	Difficulty	Working Space（Show full working）
1	The cuboid is 6 cm long, 4 cm wide and 5 cm high. Volume = ?	Basic	
2	Triangular base 8 cm, height 6 cm. Area = ?	Basic	
3	The parallelogram has a base of 12 cm and a height of 5 cm. Area = ?	Basic	
4	The base area of a cuboid is 24 cm^2 and the height is 3 cm. Volume = ?	Advanced	
5	The volume of a cuboid is 120 cm^3, its length is 6 cm, and its width is 5 cm. Seek high.	Advanced	

II.Core Knowledge ＋ Worked Examples

Knowledge point 1: Comprehensive application of volume + area formula🔴 SSPA

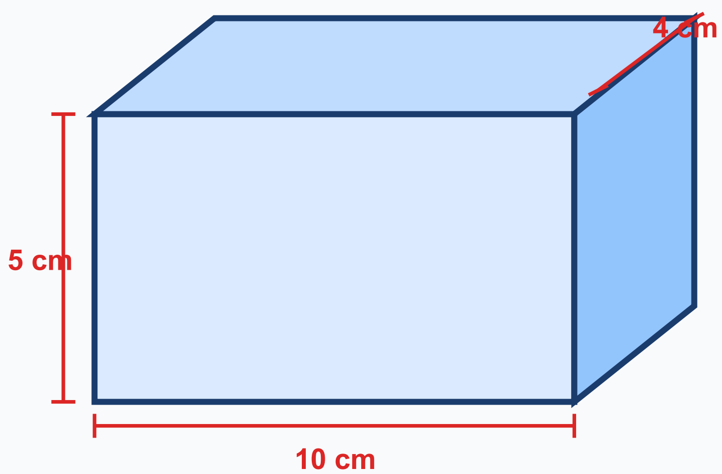
① **Volume formula (cuboid/cube):** V = length × width × height; V = base area × height

② **Area formula review:** Triangle A = (base × height) ÷ 2 · Parallelogram A = base × height · Trapezoid A = (upper base + lower base) × height ÷ 2

③ **Key distinction:** The area is the size of the "surface" (cm^2/m^2), and the volume is the size of the "space" (cm^3/m^3)

④ **Composite graphics strategy:** Divide into basic graphics → Find area/volume separately → Add or subtract

⑤ **Unit trap:** The base and height units must be consistent; the volume unit is cubic (^{3}), not square (^{2})!



V = length × width × height = 10 × 4 × 5 = 200 cm^3

Example of trap detonation (the most important demonstration in this class)

A rectangular container has a trapezoidal base (upper base 4 cm, lower base 6 cm, height 5 cm), and the height of the container is 10 cm. Find the volume of the container.

❌ Common mistakes (70% students)

$V = (4+6) \times 5 \times 10 = 500\ cm^3$

Forget about the area of a trapezoid ÷2! First find the base area = $(4+6)\times5\div2 = 25\ cm^2$, then $\times\ 10 = 250\ cm^3$

✅ Correct solution

$V = 250\ cm^3$

① Base area = $(4+6)\times5\div2 = 25\ cm^2$ ② $V = 25\times10 = 250\ cm^3$

🧠 Tip: "First determine the surface and determine the body, and distinguish the basis of the formula; divide the area of a trapezoid by two, and then multiply the volume by the height; the unit is cubed, and the area is raised to the second power."

⚠️ The most frequent error: Forget the area of the trapezoid ÷2, use (upper base + lower base) × height directly as the area. Remember that the trapezoidal formula must be divided by 2!

⚠️ The second most common error: the unit of volume is written as cm^2 instead of cm^3 - area is square and volume is cubic!

Knowledge Point - Worked Examples (write out complete steps: ①findbasearea ②findvolume ③mark unit)

#	Question	Difficulty	Working Space
Example 1	The side length of a cube is 5 cm. Find: (a) the area of a face (b) the volume	🌱	
Example 2	The cuboid is 8 cm long, 6 cm wide, and 4 cm high. Find the area and volume of the base.	🌱	
6	A rectangular parallelepiped with a square base (side length 7 cm) and height 10 cm. Find the volume.	🌿	
7	A rectangular container has a parallelogram base (base 8 cm, height 5 cm) and the height of the container is 12 cm. Find the volume.	🌿	
8	A rectangular parallelepiped has a base area of 36 cm^2 and a height of 9 cm. If the height is increased by 3 cm, how much will the volume increase?	🌳	
9	The picture below consists of two cuboids (stacked one above the other). Upper layer: 3×3×4 cm; lower layer: 5×5×6 cm. Find the total	🌿	

	volume. Tip: Calculate separately and then sum.		
10	The side length of one cube is 2 times the length of the other cube. How many times the volume of the large cube is the volume of the small cube?	🟢	

Knowledge Point 2: Application of Fractions in Volume 🔴 SSPA Required Exam

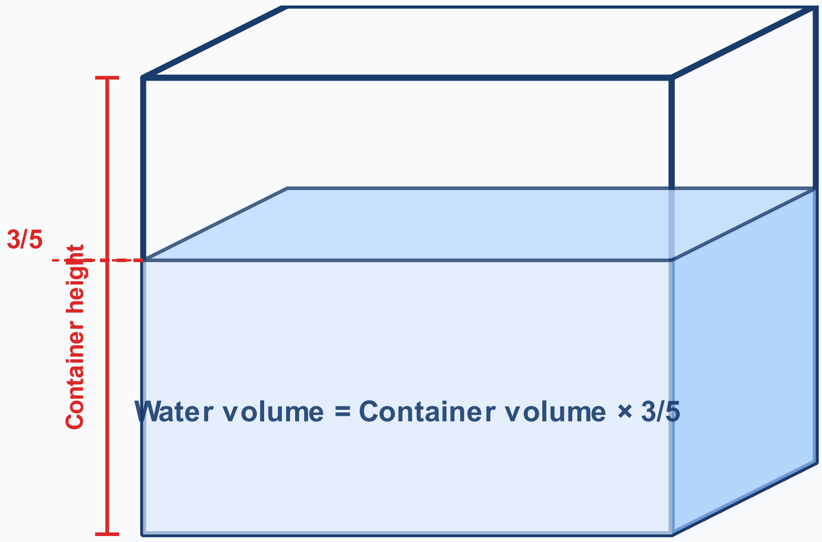
Four ways to consider fractions in volume:

① **Side length contains fractions:** Length = $2\frac{1}{2}$ cm → Convert to false fraction and multiply: V = $\frac{5}{2}$ Container volume

② **Fraction of volume:** The container is packed $\frac{3}{4}$ of water → volume of water = volume of container × $\frac{3}{4}$

③ **Find the volume of the parts and find the whole:** known $\frac{2}{5}$ The volume is 40 cm³ → whole = 40 ÷ $\frac{2}{5}$ = 100 cm³

④ **Scaling:** Length enlargement $1\frac{1}{2}$ times → volume enlargement $(\frac{3}{2})^3 = \frac{27}{8}$ times



① **Side length includes fraction** Mixed fractions→improper fractions and then multiply to find the volume	② **Find the fraction** Volume × Fraction = Partial Volume	③ **Rounding the known parts** Part ÷ Fraction = Whole Volume	④ **scaling** Side length × kVolume × k³

example

Example 3: A cuboid is $2\frac{1}{2}$ cm long, 4 cm wide, and 3 cm high. Find the volume.

example

Example 4: A water tank has a volume of 80 L and is filled with $\frac{3}{5}$ water. What is the volume of water in L?

❌ Common mistakes

$$2\frac{1}{2} \times 4 \times 3 = 2.5 \times 12 = 30$$

After converting mixed numbers to improper fractions, the numerator and denominator are processed separately: 52×4×3 = 30 ✓ (This question happens to be correct), but it is easy to get confused in complicated situations!

✅ Correct approach

$$V = \frac{5}{2} \times 4 \times 3 = 30\ \text{cm}^3$$

① $2\frac{1}{2} = \frac{5}{2}$ ② $\frac{5}{2} \times 4 \times 3 = \frac{60}{2} = 30$

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
11	The side length of the cube is $1\frac{1}{2}$ cm. Find the volume. (Answer in fractions)	🌿	
12	A water tank has a volume of 60 L and contains $\frac{2}{3}$ water. How many liters of water is there?	🌿	
13	A fish tank $\frac{3}{4}$ has 45 L of water. What is the volume of the entire fish tank in L?	🌳	
14	The length of the cuboid is $\frac{3}{2}$ m, the width $\frac{2}{3}$ m, and the height is 2 m. Find the volume. (Answer with m^3)	🌳	
15	After the side length of the cube is enlarged by $1\frac{1}{3}$ times, how many times is the new volume the original? (Answer in fractions)	⛰️	

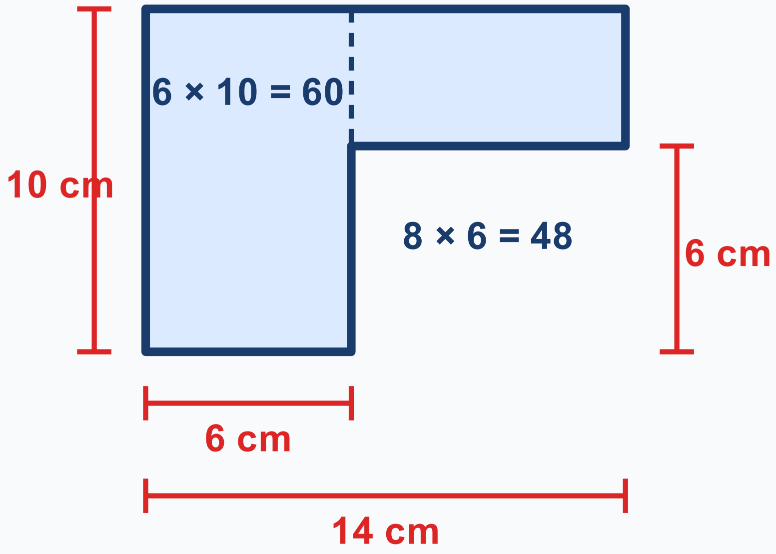
Knowledge Point 3: Comprehensive Application—Volume + Area + Fraction Three-in-One 🔴 SSPA Advanced

Strategies for solving cross-topic comprehensive questions (three-step method for reading questions):

① **Circle all the numbers and units in the question**- Clarify which are areas, which are volumes, and which are fractions

② **Judge the order of operations**- First find the area (base area) → then find the volume → finally deal with the fraction ratio

③ **Check the consistency of the units**——m and cm cannot be multiplied directly! Convert to the same unit first



Total area = 60 + 48 = 108 cm²

example

Example 5: A rectangular water tank with a rectangular base (1 m long, 50 cm wide) and 80 cm high. The water tank contains $\frac{3}{5}$ water. Find the volume of water (answer in L, 1 L = 1000 cm^3).

example

Example 6: The side length of cube A is 6 cm, and the side length of cube B is $\frac{2}{3}$ of A. What fraction of the volume of B is A?

🧠 Tip: "Unify the units first, then the area and then the volume; multiply the fractions last, and avoid all traps."

⚠ High frequency trap: m and cm are mixed up - 1 m = 100 cm, but 1 m^2 = 10000 cm^2, 1 m^3 = 1,000,000 cm^3!

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space

16	The base area of a cuboid is 25 cm² and the height is 8 cm. What is the volume $\frac{3}{4}$ in cm³?	🌿	
17	A fish tank is 50 cm long, 40 cm wide, and 30 cm high. (a) Find the volume of the fish tank (b) If water is filled to the height of |||SEP|||, what is the volume of water in L? $\frac{4}{5}$ Height, what is the volume of water in L?	🌳	
18	The side length of a cube water tank is 20 cm. Existing water 4000 cm³. What fraction of the tank's volume does water occupy? (Answer with the simplest fraction)	🌳	
19	The cuboid is 1.5 m long, 0.8 m wide, and 0.6 m high. (a) Volume = ? m³ (b) If the space of $\frac{5}{8}$ has been used, how much m³ is the remaining space?	🌳	
20	A cylinder has a base area of 78.5 cm² and a height of 10 cm. If the height is reduced by |||SEP|||, what is the new volume? (Hint: new high = 10 × (1 − $\frac{1}{5}$, what is the new volume? (Hint: new high = 10 × (1 − $\frac{1}{5}$))	⛰️	

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
21	The cuboid is 10 cm long, 5 cm wide, and 4 cm high. Find the volume.	🌱	
22	The side length of the cube is 8 cm. Find the volume.	🌱	
23	Triangular base 12 cm, height 9 cm. Find the area.	🌱	
24	The upper base of the trapezoid is 4 cm, the lower base is 8 cm, and the height is 5 cm. Find the area.	🌱	
25	The volume of a cuboid is 72 cm³ and the base area is 12 cm². Seek high.	🌱	

Advanced layer (total 5 questions, 🧗🚀 choose do)

#	Question	Difficulty	Working Space
26	A cuboid has a square base (sides 6 cm) and a height 15 cm. Find the volume.	🌿	

27	Water tank volume 120 L, used |||SEP|||. How much L space is left? $\frac{5}{6}$. How much L space is left?	🌿	
28	The side length of the cube is $1\frac{1}{3}$ cm. Find the volume. (Answer with improper fractions)	🌿	
29	Cuboid A: length 6 cm, width 4 cm, height 5 cm. Cuboid B: All sides are twice as long as A. How many times the volume of B is A?	🌳	
30	A combined solid: the lower cuboid is 10×8×5 cm, and the upper cube with side length 4 cm is placed in the center. Find the total volume.	🌳	

🌳 challenge layer (total 3 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
31	A rectangular container has an interior length of 30 cm, a width of 20 cm, and a height of 15 cm. After placing a stone, the water level rises by 2 cm. What is the volume of the stone in cm^3? Tips: Drainage method - stone volume = bottom area × water level rise height	🌳	
32	The side length of the cube is $\frac{3}{4}$ m. Find (a) the area of a surface (answer in m^2) (b) the volume (answer in m^3) (c) how many times the volume is the area of the base?	🌳	
33	A trapezoidal cylinder: the base is trapezoidal (upper base 5 cm, lower base 9 cm, height 4 cm), and the height of the cylinder is 10 cm. Find the volume. Tips: V = base area × height = trapezoid area × cylinder height	⛰️	

IV.applicationquestionspecial topic training

#	Question	Difficulty	Working Space
34	A rectangular pool is 8 m long, 5 m wide, and 2 m deep. (a) What is the volume of the pool in m^3? (b) If the water depth is only |||SEP|||, what is the volume of water in m^3? $\frac{3}{4}$, what is the volume of water m^3?	🌿	
35	A rectangular container is 12 m long, 2.5 m wide, and 2.8 m high. (a) What is the volume of the container in m^3? (b) The loaded goods occupy |||SEP|||, how many m^3 of goods can be loaded in the remaining space? $\frac{5}{7}$, how many m^3 of goods can be loaded in the remaining space?	🌳	

36	A cube water tank has a side length of 40 cm. 8000 cm^3 of water is injected every minute. (a) What is the total volume of the tank in cm^3? (b) How many minutes does it take to fill the water tank? (c) After 5 minutes, what fraction of the water will be in the tank?		
37	A rectangular fish tank is 60 cm long, 30 cm wide, and 40 cm high. After first pouring 36 L of water and adding some stones, the water level rose by 5 cm. (a) 36 L What is the height of the water in the tank in cm? (1 L = 1000 cm^3) (b) What is the total volume of the stone in cm^3? (c) What is the final water depth in cm?		

V. Ultimate challenge area

#	Question	Difficulty	Working Space
1	The length of a cuboid is 2 times the width, and the breadth is $1\frac{1}{2}$ times the height. If the height is 4 cm, find: (a) What is the width? (b) What is the length? (c) What is the volume? (d) What is the surface area? (Surface area = 2×(length×width + length×height + width×height))		
2	A cube has a side length of 10 cm. A small cube with a side length of 2 cm is dug out in the center of each of its six faces (digging through to the opposite side). Find the volume of the remaining solid. Tips: Note that the perforations in the three directions will overlap, be careful not to deduct repeatedly!		
3	A trapezoidal cylindrical pool: the bottom is trapezoidal (upper bottom 3 m, lower bottom 7 m, height 4 m), and the pool depth is 2 m. 32 m^3 of water available. (a) What is the total volume of the pool in m^3? (b) What fraction of the total pool is the existing water? (Answer in simplest fractions) (c) How many m^3 of water must be injected to fill it completely?		

Cross-topic integration: volume + area + fraction, a universal problem-solving framework essential for score examinations

When encountering comprehensive questions, deal with them in the following order:

① **Read the questions:** Circle all the numbers, mark the units (cm/m/L/cm^3/m^3), and find out "a fraction"

② **Unify the units:** Unify all lengths into the same unit (all cm or all m)

③ **Area first, then volume:** If you need the base area → use the corresponding formula

(rectangle/triangle/trapezoid/parallelogram)
④ **Processing fractions:**Volume × fraction = volume of part; part ÷ fraction = whole
⑤ **Check the answer:**Are the units correct? Are fractions reduced? Is the answer reasonable?

VI.Class afterhomework

Basic must-do questions (total 5 questions, mustShow full working)

#	Question	Difficulty	Working Space
H1	The cuboid is 7 cm long, 5 cm wide, and 3 cm high. Find the volume.	🌱	
H2	The side length of the cube is 9 cm. Find the volume.	🌱	
H3	A cuboid has a base area of 28 cm² and a height of 6 cm. Find the volume.	🌱	
H4	Water tank volume 90 L. Installed $\frac{2}{5}$ water. How many liters of water is there?	🌿	
H5	A cuboid is 1.5 m long, 0.8 m wide, and 0.5 m high. Find the volume (answer in m³).	🌿	

Advanced choose doquestion (total 3 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H6	The cuboid is $2\frac{1}{2}$ cm long, 4 cm wide and 3 cm high. Find the volume.	🌿	
H7	A cube water tank has a side length of 25 cm. Pour 8 L of water first, then add the stones, and the water level rises by 3 cm. What is the volume of the stone in cm³? (1 L = 1000 cm³)	🌳	
H8	Cuboid A: 6×4×5 cm. Cuboid B: All side lengths are $1\frac{1}{2}$ times that of A. How many times the volume of B is A? (Answer in fractions)	⛰️	

VII. The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🌱basic questions independently
☐ I can challenge 🌿advanced questions
☐ I remember the formula

🎯 Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall
☐ Solve 🌱basic questions independently (100% correct)
☐ Challenge 🌿Advanced questions (80%+ correct)
☐ Explain the lesson formula to classmates

✅ 本堂自我檢查（完成後打剔）

☐ 我識得分辨每個知識點嘅陷阱
☐ 我能夠獨立完成🌱基礎題
☐ 我能夠挑戰🌿進階題
☐ 我記得住口訣

#	common error	Correct Approach
1	**Forget the area of the trapezoid ÷2**(upper base + lower base) × height, not the area of the trapezoid!	Trapezoidal area = (upper base + lower base) × height ÷ 2, you must also divide 2 when using it to calculate the base area.
2	**The volume unit is written incorrectly |||SEP|||: cm^3 is written as cm^2**: cm^3 is written as cm^2	Area → square (^{2}), volume → cubic (^{3}). Check the index before each answer
3	**m/cm conversion error |||SEP|||: 1m=100cm but $1m^3 \neq 100cm^3$**: 1m=100cm but $1m^3 \neq 100cm^3$	$1\ m^3 = 100 \times 100 \times 100 = 1{,}000{,}000\ cm^3$. Three-dimensional conversion requires conversion in each dimension
4	**The fractions are not reduced after multiplication**：$\frac{6}{12}$ Take it directly as the answer	The last step is to check whether the numerator and denominator have common factors, which must be reduced to the simplest
5	**"Part ÷ Fraction = Whole" is reversed |||SEP|||: Use multiplication instead of division**:Use multiplication instead of division	Given that $\frac{3}{5}$ is 60 → overall = 60 ÷ $\frac{3}{5}$ = 100 (not 60 × $\frac{3}{5}$)
6	**The side length is enlarged k times → the volume is enlarged k times**(Wrong!)	The side length is enlarged k times → the volume is enlarged k^3 times (cubic!)
7	**Misunderstanding of drainage method |||SEP|||: Thinking that the rising height of the water level = the height of the stone**: Thinking that the rising height of the water level = the height of the stone	The volume of the stone = the bottom area of the container × the rising height of the water level (not the height of the stone)

🏠 家長角落 · Parent Corner

今日學咗咩？ 小朋友學咗「**Comprehensive—Volume + Area + Fraction**」。
最易錯嘅 3 個陷阱： 留意老師圈出嘅陷阱題目
你可以問小朋友：「你可唔可以解釋「Comprehensive—Volume + Area + Fraction」嘅最重要口訣俾我聽？」
溫馨提示：唔需要識教數學 — 只需要問小朋友「點解咁計？」同「有冇檢查陷阱？」就夠。

📝 **教師參考**：熱身題答案 → 1)___ 2)___ 3)___ 4)___ 5)___ | 課堂練習重點關注題號：🌳挑戰層 17-21